

HARIN DULNETH

SOFTWARE ENGINEER INTERN

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SUMMARY

Computer Science undergraduate specializing in AI-driven, production-grade systems. Experienced in building scalable backend services with .NET and FastAPI, and integrating machine learning and LLM solutions using PyTorch and TensorFlow. Skilled in developing end-to-end ML pipelines, RESTful architectures and data-driven applications with PostgreSQL, MongoDB and Redis. Familiar with Docker and CI/CD workflows, ensuring reliable and maintainable deployments. Passionate about leveraging AI to build practical, user-centric and high-performance software solutions.

EDUCATION

Bsc. in Computer Science (Hons)

2023 - Present

University of Kelaniya, Sri Lanka

Specialize in Artificial Intelligence (Current GPA - 3.82)

G.C.E Advanced Level - 1A 2Bs

2019 - 2022

Physical Science Stream

Isipathana College, Colombo 05

G.C.E Ordinary Level - 8As 1B

2008 - 2018

Isipathana College, Colombo 05

CORE COMPETENCIES

- **Programming Languages** - Python | C | C# | Java
JavaScript | TypeScript
- **Backend** - FastAPI | Flask | Node.js | .NET
Spring Boot | Express.js
- **Databases** - SQL | PostgreSQL | Redis
MongoDB | Supabase
- **Machine Learning / AI** - TensorFlow | PyTorch
Hugging Face | MLOps | LLM
- **Frontend** - React | Next.js | Tailwind
- **Other** - Git | Docker | CI/CD | Linux | Google Colab
Figma | AWS

PROJECTS

Trowd - Real-Time Transit Intelligence Platform | [Project](#)

Technical Stack - React (TypeScript), FastAPI, Node.js, PostgreSQL, Redis, Socket.IO, LightGBM, XGBoost

- Built a full-stack bus tracking system with ML-based ETA forecasting via a fusion engine using confidence-weighted blending of real-time telemetry and historical segment data.
- Engineered a crowd-sourced arrival confirmation pipeline with GPS clustering, temporal consistency checks, and dynamic reputation-based feature weighting.
- Implemented a self-improving feedback loop for continuous model retraining and a low-latency pub/sub architecture for live bus location, ETA, and occupancy updates.

CuraNex - AI Demand Forecasting & Inventory Optimization Platform | [Project](#)

Technical Stack - Python, React (TypeScript), TensorFlow, PyTorch, XGBoost, LightGBM

- Led development of an AI-driven demand forecasting platform for AITHON 2026 (MVP phase) organized by Hemas Holdings PLC, predicting stock requirements and detecting inventory anomalies to optimize supply chain efficiency.
- Built a scalable ML pipeline integrating deep sequence models (TFT, NBEATS) and gradient boosting with dynamic recursive feature engineering.
- Designed an interactive analytics dashboard with forecast visualisation, anomaly alerts, and proactive inventory recommendations to support decision-making.

Portfolio Website With AI Assistant | [Live Website](#) | [Project](#)

Technical Stack – Next.js 15, Tailwind CSS, Hugging Face, Upstash Vector DB, Groq (LLaMA 3.1), Redis

- Engineered a portfolio website with an AI assistant using Retrieval-Augmented Generation (RAG) to answer user queries based on indexed resume and GitHub data.
- Built a serverless AI pipeline integrating embeddings, vector search, and Redis-based session memory with automatic conversation summarization.
- Designed a highly interactive UI featuring WebGL shaders, Spline 3D elements, and dynamic visual effects for an engaging user experience.

CodeMart – Digital Project Marketplace | [Project](#)

Technical Stack – React (TypeScript), ASP.NET Core, Entity Framework Core, PostgreSQL (Supabase), Docker, Stripe API

- Developed a full-stack digital marketplace for buying and selling software projects with secure authentication, payment integration, and role-based access control.
- Built a RESTful backend supporting project listings, order management, reviews, and transaction workflows.
- Designed a responsive and visually rich frontend using Three.js, GSAP animations, and modern UI components to deliver a premium user experience.

Whisper — Full-Stack Secure Messaging App | [Project](#)

Technical Stack – React (TypeScript), Java, Spring Boot, PostgreSQL, WebSockets, JWT, Tailwind CSS, Cloudinary

- Developed a full-stack secure web chat application for private communication, featuring OTP-based registration and real-time instant messaging.
- Built a robust backend supporting real-time messaging workflows, secure authentication, profile management, and SMTP Gmail integration for OTP delivery.
- Designed a responsive and user-friendly frontend using React and Tailwind CSS, featuring a clean modular architecture and seamless contact management UX.

CERTIFICATES

- Deep Learning Specialization | DeepLearning.AI | Credential ID: [PA5O7P3AQ85C](#)
- AI & ML Engineering - Professional Certificate | Microsoft | Credential ID: [DZ6XR637YI5Q](#)
- AWS Cloud Technical Essentials | AWS | Credential ID: [FDLMXF9TS4GQ](#)
- AI/ML Engineer - Stage 2 | SLIIT Faculty of Computing | Credential ID: [txqnjcawyu](#)
- IBM Cloud Essentials | IBM | Credential ID: [ededb5b52cb14d2e97a793e6a9d9ca6f](#)

ACHIEVEMENTS

- **KelaniXtreme - 1st Place (IEEE Student Branch University of Kelaniya)**
- **AITHON 2026 - Top 30 Finalist & MVP Phase (Hemas Holdings PLC)**
- **Selected for Presentation (FCTSRs 2025) – “Nexus: Centralized Platform for Planning and Managing Projects”**
- **Participated in SpiritX - Inter University Development Competition (MoraSpirit)**
- **Participated in IX'25 - Designathon (IEEE Student Branch of IIT)**

VOLUNTEERING

- Team Member - IEEE EPICS Global, Development of a Digital Appointment System for Homagama Base Hospital Eye Clinic
- Member - IEEE Sri Lanka Section (2024/25)
- Member - Computer Science Student Association (2023 - Present)
- Member - Free and Open-Source Software Community (2024 - Present)
- Rugby - 2 Runners up (Inter Faculty Tournament 2025)